

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

16th March 2017

Subject CT4 – Models

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 3. Mark allocations are shown in brackets.*
- 4. Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) i) Explain what is meant by a stochastic model and give two advantages stochastic models have over deterministic models. (3)

ii) A company is studying the health records of its longest serving employee in order to improve its provision for health insurance. Let $X(t) = H$ if the employee is healthy at time t , $X(t) = S$ otherwise. The available information includes the value of $X(t)$ for all $0 \leq t \leq T$. The company chooses to model X as a two-state time-homogeneous Markov jump process with transition rates $\sigma_{HS} = \sigma, \sigma_{SH} = \rho$.

Having fitted the model the company discovers that the observed distribution of holding times in state H does not fit the predictions of the time homogeneous Markov model; in particular, the mean holding time in state H between visits to state S appears to be decreasing with t .

Explain why the time homogeneous model could still be used if the company is large and has a roughly constant age profile. (3)

iii) X_n is a stochastic process with a discrete state space and a discrete time set. Show that if non-overlapping increments of this process are independent, then the process satisfies the Markov property. (3)
[9]

Q. 2) An investigation was undertaken into the mortality of newly born mosquitoes. The investigation reveals that for first 30 days the newly born mosquitoes have a constant force of mortality of 0.03.

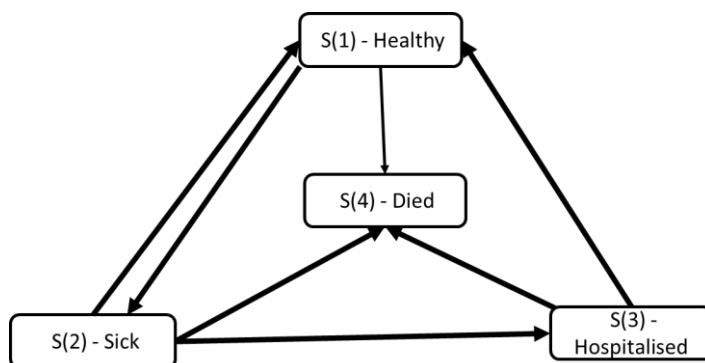
Thereafter, force of mortality increases according to the formula:

$$\mu_{30+x} = -0.5 + \lambda x, \text{ where } x \text{ denotes the number of days after day 30.}$$

i) Calculate the probability that a newly born mosquito will survive for at least 15 days. (2)

ii) Calculate the parameter λ , if it is known that 95 percent of mosquitoes are expected to have died within 90 days of age. (5)
[7]

Q. 3) Consider the following multiple-state model in which $S(t)$, the state occupied at time t by a life initially aged x , is assumed to follow a continuous-time Markov process. The States are discrete and can be identified as 1- Healthy, 2- Sick, 3- Hospitalised and 4- Died.



Let μ_{x+t}^{ij} denote the force of transition at age $x + t$ ($t \geq 0$) from State i to State j , and let

$${}_t p_x^{ij} = P(S(t) = j | S(0) = i)$$

- i) Derive from first principles the forward differential equation:

$$\frac{\partial}{\partial t} {}_t p_x^{21} = {}_t p_x^{22} \mu_{x+t}^{21} + {}_t p_x^{23} \mu_{x+t}^{31} - {}_t p_x^{21} (\mu_{x+t}^{12} + \mu_{x+t}^{14})$$

State all the assumptions that you make.

(5)

- ii) Write down forward differential equations for ${}_t p_x^{23}$ and ${}_t p_x^{32}$ with suitable general reasoning.

(3)

[8]

- Q. 4)** A Life Insurance company runs a training course for sales trainees each year. The course lasts for 30 days, but the examination which enables someone to qualify as a sales agent can be sat any day during the course. In November, 2016 there were 12 participants who started the training course. The following table has been compiled to show the day each trainee qualified as sales agent or the day each trainee who did not qualify left the course.

Candidate	Day qualified	Day left without qualifying
A		10
B		20
C	18	
D	13	
E		30
F	2	
G		18
H	13	
I		30
J	15	
K		6
L	24	

- i) Define Type I and Type II censoring? (2)
- ii) Describe the type of censoring present in the above data? (2)
- iii) Calculate the Kaplan-Meier estimate of the non-qualification function. (5)
- iv) State the use and limitations of Kaplan Meier estimate of survival function? (2)

[11]

- Q. 5)** An Actuary has 4 umbrellas, some at home, and some in the office. He keeps moving between home and office and takes an umbrella with him only if it rains. If it does not rain, he leaves the umbrella behind (at home or in the office). It may happen that all umbrellas are in one place and if he is at the other end and if it starts raining he will get wet (if it rains, he compulsorily takes an umbrella if it is available, else he must move out without an umbrella and reach the other side after getting wet). The probability of raining is p .

- i) Consider the number of umbrellas in the place where the actuary currently is to be the state space of a Markov chain, draw the diagram of the process, and simplify it, if possible (3)
 - ii) Write the transition matrix for this chain. (2)
 - iii) Determine the stationary distribution of the process (4)
 - iv) What is the probability that he gets wet? (2)
 - v) Current estimates show that $p = 0.6$. How many umbrellas should he have so that, if he follows the strategy above, the probability of his getting wet is less than 0.01? Provide suitable comment. (4)
- [15]**

Q. 6) An investigation took place into the effect of a new treatment on the survival of cancer patients. Two groups of patients were identified. One group was given the new treatment and the other an existing treatment.

The following model was considered:

$$h_i(t) = h_0(t) * \exp(\beta^* z)$$

where: $h_i(t)$ is the hazard of death at time t , where t is the time since the start of treatment
 $h_0(t)$ is the baseline hazard at time t

z is a vector of covariates such that:

z_1 = sex (a categorical variable with 0 = female, 1 = male)

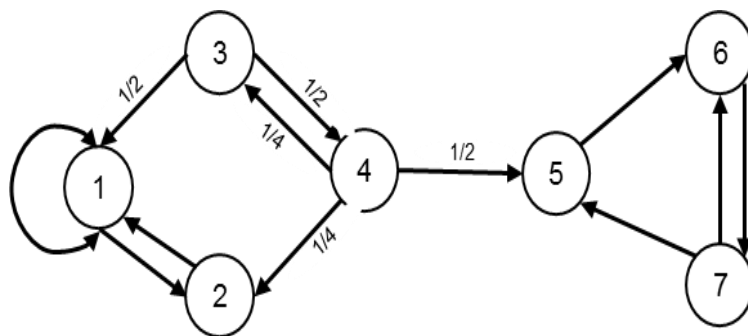
z_2 = treatment (a categorical variable with 0 = existing treatment, 1 = new treatment)

and β is a vector of parameters, (β_1, β_2)

The results of the investigation showed that, if the model is correct:

- I. The risk of death for a male patient is 5% more than that of a female patient; and
 - II. The risk of death for a patient given the existing treatment is 10% more than that for a patient given the new treatment
- i) State the group of patients to which the baseline hazard applies? (1)
 - ii) Prove that the risk of death for a male patient who has been given the new treatment is less than that for a female patient given the existing treatment. (4)
 - iii) Also, determine the probability that a male patient will die within 3 years of receiving the new treatment, if the median survival time for female patient whom existing treatment given is 3 years. (3)
- [8]**

Q. 7) Consider the Markov chain in the following diagram.



Identify two sets of states $R1=\{1,2\}$, and $R2=\{5,6,7\}$.

- i) Draw the reduced diagram showing $R1$ and $R2$ explicitly and determine if the process is Irreducible. (2)
- ii) What types of states $R1$ and $R2$ are? (1)
- iii) Assuming $X_0=3$, find the probability that the chain gets absorbed in $R1$. (3)
- iv) Again assume $X_0=3$. We would like to find the expected time (number of steps) until the chain gets absorbed in $R1$ or $R2$. More specifically, let T be the absorption time, i.e., the first time the chain visits a state in $R1$ or $R2$. We would like to find $E[T|X_0=3]$. (3)

[9]

Q. 8) i) State the principle of correspondence as it applies to mortality rates. (1)

The experience data of a life insurance company is given below:

Age last birthday	Number of policies in force as on date for each age last birthday			
	01-01-2014	01-01-2015	01-07-2015	01-01-2016
55	50000	55000	60000	62000
56	49000	54000	58000	59000
57	53000	54000	56000	58000

Number of deaths classified by age next birthday and calendar year

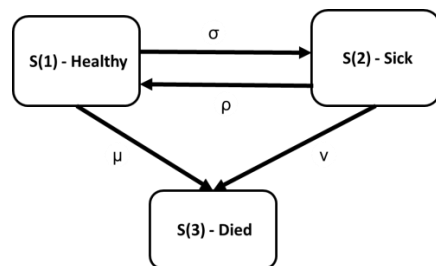
Age next birthday	2014	2015
55	1100	1200
56	1150	1225
57	1300	1325

- ii) Estimate, the force of mortality at age 56 next birthday for the period 1st January 2014 to 1st January 2016. (5)

Further, the age definition of policies in-force was wrongly intimated and the actual age captured was age next birthday.

- iii) Discuss the implication if the results arrived in (ii) above were used for pricing a term insurance product without correcting the age definition of inforce policies? (3)
[9]

Q. 9) i) In a municipal hospital, records of sickness and deaths are maintained. Consider Healthy(1) –Sick(2) –Died(3) model with constant transition rates.



- a) Write down the likelihood of the data in the investigation in terms of the transition rates and the waiting times in the Healthy and Sick states, under the assumption that the transition rates are constant. (2)

- b) Derive the maximum likelihood estimator of the transition rate from Sick to Dead. (3)

c) Following data is given:

Total time for healthy state:	3000
Total time for Sick state:	1600
Number of transitions from Healthy to sick :	300
Number of transition from sick to dead:	700

Estimate the transition rate from Sick to dead and 95 percent confidence interval around this transition rate. (3)

- ii) Now consider the reduced form of the above model with only two states, Alive and Dead. Such models are used to investigate mortality.

- a) If hazard rate is estimated to be 0.2 in certain experiment, calculate expected value of waiting time for i th life $E(V_i)$ for an investigation period of 1 year. If Total Number of lives under estimation were 2000, determine the total expected waiting time $E(V)$. (2)

- b) Mathematically write the equation for distribution of Actual Hazard rate and determine the probability that actual hazard rate is greater than 0.21. (2)
[12]

- Q. 10) i)** List the methods of graduation. Describe how smoothness is ensured when mortality rates are graduated using each method? (3)
- ii)** Explain why a mortality experience would need to be graduated. (2)

An actuary has conducted investigations into the mortality of the following classes of lives:

- I.** Members covered under PMJJY (Pradhan Mantri Jeevan Jyoti Bima Yojana)
- II.** Female population of a large developing country
- III.** Patients suffering from brain cancer in a small province.

The actuary wishes to graduate the crude rates for each of the above classes of lives

- iii)** State an appropriate method of graduation with rationale for each of the three classes of lives and, for each class, briefly explain your choice. (3)

A large life insurance company has graduated the mortality experience of a business portfolio. The original data and the graduated rates are as follows.

Age	Exposed to risk	Number of deaths	Graduated Rates
50	1280	4	0.00230
51	2030	5	0.00262
52	1950	11	0.00296
53	2160	7	0.00331
54	2480	10	0.0037
55	1455	7	0.00415
56	2100	11	0.00463
57	1865	17	0.00518
58	1990	16	0.00578
59	1725	9	0.00645

- iv)** Test the graduation for overall goodness of fit. (4)
- [12]**
